

ASSIGNMENT 4.4

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseat – Step by Step Process

1 Prepare & Power Off  1. Shut down the laptop completely. 2. Disconnect the charger. 3. Place the laptop on a clean, flat surface.	2 Open the Back Panel  1. Remove the screws from the back panel. 2. Carefully lift off the panel using a plastic tool or your hands.	3 Locate the RAM Slots  1. Find the RAM slots on the motherboard. 2. In my laptop, there are 2 slots (one is occupied, the other is empty).	4 Remove the RAM Module  1. Gently push the side clips outward (left and right). 2. The RAM module will pop up. 3. Hold it by the edges and remove it.	5 RAM Module Removed  1. The module is now completely removed. 2. Keep it in a safe place...
6 Re-insert the RAM Module  1. Align the notch on the RAM with the slot. 2. Insert the module at an angle (~30°). 3. Press it down gently until the side clips lock it in place.	7 Check the Fit  1. Make sure the module is fully seated. 2. The clips should be locked and hold the RAM firmly.	8 Close the Laptop  1. Put the back panel in place. 2. Tighten all the screws.	9 Power On & Test  1. Connect the charger. 2. Turn on the laptop. 3. Check if the system boots properly and the RAM is recognized.	Issues Observed  During the process: - No major issues faced. - RAM module was easy to remove and re-insert. - The clips were a bit tight, so I used gentle pressure.  After re-inserting: - Laptop booted normally. - RAM was detected in BIOS (checked by pressing F2 at startup). - System is running smoothly without any error.  Conclusion: The RAM reseating process went smoothly. No issues were observed, and the laptop is working fine.

Before you start

- Turn off the PC and unplug the power cable.
- Press and hold the power button for 10 seconds.
- Use a screwdriver and work on a clean table.
- Touch a metal part of the case to remove static.

Steps to reseat RAM

1. Open the PC case or laptop back cover.
2. Find the RAM sticks next to the CPU fan.
3. Push the clips on both sides of the RAM to release it.
4. Take the RAM out by holding the edges (do not touch the gold pins).
5. Wipe the gold pins gently with a dry, soft cloth.
6. Match the notch on the RAM with the slot and push it in firmly.
7. Listen for a click. The clips should lock by themselves.
8. Close the case, plug in power, and turn on the PC.

Common issues and what to do

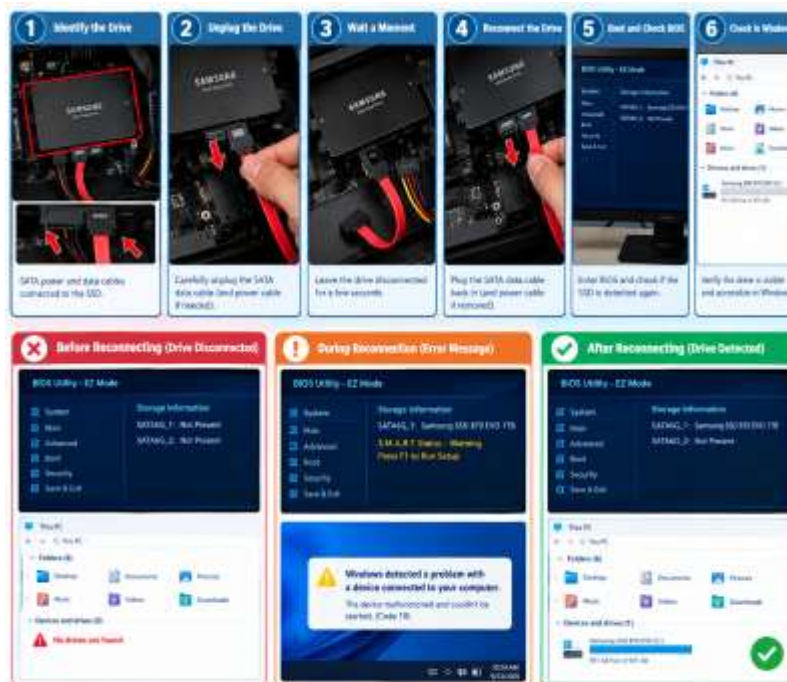
- PC does not start: check if RAM is fully pushed in and in the right direction.
- Beep sound or memory error on screen: try one RAM stick at a time.
- BIOS shows less RAM: reseal the RAM again and check in BIOS.
- Always ground yourself before touching RAM.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

1. Check symptoms: no display, strange colors/lines, game crashes, black screen, or driver errors.
2. Check cables and power: make sure GPU is tight in the slot and all power cables are connected.
3. Test monitor and cable: try another cable or monitor. If CPU has video output, try that too.
4. Update or reinstall GPU driver: remove old driver, then install the latest driver from the company website.
5. Run a GPU test: use a stress test app and watch temperatures. If problems continue, GPU may be faulty.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.



When drive is unplugged

- BIOS: drive not shown in SATA or NVMe list.
- If it was the main drive: “No boot device” or “Operating system not found”.
- Windows: drive letter missing in File Explorer.
- Disk Management: disk missing or shown as “Unknown”.

When drive is plugged back in

- BIOS: drive appears again in SATA/NVMe list.
- Windows: drive letter shows up again (for data drives).
- If cable is loose: drive may come and go, or show “Disk read error”.
- If NVMe is not tight: BIOS may still not see it. Reseat and tighten the screw.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

First component to check

First check: RAM

Why RAM first?

Bad RAM can corrupt data and make the PC restart suddenly. RAM tests are easy and fast.

Steps to check RAM

1. Run Windows Memory Diagnostic: search “Windows Memory Diagnostic”, run it, and restart. Check results in Event Viewer.
2. Run MemTest86: make a bootable USB, boot from it, and run at least 4 passes. Any error means bad RAM.
3. Test one stick at a time: if you have two or more sticks, test each one alone.
4. Check BIOS: if you use XMP/overclock, turn it off and use default RAM speed.
5. Reseat RAM: power off, take out RAM, and put it back in firmly. Then test again.

If RAM is OK

Next check GPU (drivers, artifacts, stress test), storage (SMART health, errors), power supply, and temperatures.